Noise construction and finite-size inference in temporally correlated ballistic deposition

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Temporal memory changes interface roughening, but comparing numerical results requires specifying both the driving process and the observation protocol. We examine a shared-innovation autoregressive (AR) sum and two Gaussian spectral filters in parallel-update ballistic deposition. Validation of the final binary space–time field detects an axis-assignment fault that retains correct one-dimensional sequences. Exact AR covariances retain cross terms between modes driven by common innovations. Matched campaigns with 64 independent runs per cell and $L = 32$–512 show increasing finite-size late-time width slopes for all three inputs. At $\theta = 0.25$, the slopes are 0.717 ± 0.008 for shared AR and 0.678 ± 0.007 for the fGn-spectrum filter. A paired bootstrap does not resolve a slope difference between the Gaussian filters at any of the seven tested parameters. Additional controls compare nested observation windows under one fixed spectral record, separating duration sensitivity from changes in its cutoff. The nearest-neighbour increment distribution reduces exactly to the flat-pair fraction, so its Hellinger distance describes local probability changes rather than independent spatial information. These results provide a controlled finite-window comparison and specify the input and inference checks needed before drawing asymptotic scaling conclusions.

I. INTRODUCTION

The Kardar–Parisi–Zhang (KPZ) equation describes the competition between smoothing, lateral growth and stochastic driving in a fluctuating interface [1, 2],

$$\partial_t h = \nu \nabla^2 h + \frac{\lambda}{2}(\nabla h)^2 + \eta. \tag{1}$$

For short-range noise in one dimension, the roughness and dynamic exponents are $\alpha = 1/2$ and $z = 3/2$. Their stability when the noise retains temporal memory remains a problem in nonequilibrium statistical physics.

A conventional continuum description introduces the low-frequency spectrum [3, 4]

$$D(\omega, k) = D_0 + D_\theta |\omega|^{-2\theta}, \qquad 0 < \theta < 1/2, \tag{2}$$

with short-range spatial correlations and a long-time covariance contribution proportional to $|t - t'|^{2\theta-1}$. The exponent, relative amplitudes and short-time regularisation are distinct properties. A discrete fGn covariance specifies more than this long-time power. Matching its complete shape does not by itself identify a continuum universality class.

Thresholds in this literature also concern different physical questions. The nonperturbative renormalisation-group treatment finds a short-range stability boundary at $\theta = 1/6$ in one dimension [4]. Alés and López identify a boundary near 0.25 from the separation of global and spectral roughness exponents and the appearance of anomalous, faceted growth [5]. Height-distribution markers and emergence diagnostics probe other observables [6, 7]. These definitions are not interchangeable estimates of one critical parameter. Model and discretisation dependence under temporally correlated driving has also been emphasised [8].

A numerical comparison must establish which random field reaches the growth rule. An array assembly can retain correctly filtered sequences while changing whether their correlations act in space or time. A second issue arises in inference: a finite-ensemble distance has a nonzero null floor, and its first detectable deviation depends on sampling and system size. Neither issue is resolved by fitting one exponent to an entire finite-lag covariance curve.

The question addressed here is which finite-window comparisons remain interpretable when the complete input process and its implementation are specified. We validate the assembled noise field, retain the cross terms in the shared-innovation AR covariance, and compare three inputs on a matched finite-size grid. A paired analysis of the Gaussian filters uses their common random streams rather than treating their agreement as independent replication. Additional duration controls keep the spectral record fixed while extending the observation window. Together these checks separate implementation validity, differences between specified processes, and sensitivity to observation time.

Hellinger distance and novelty detection have already been used in this setting [7], and microscopic sensitivity has been discussed [8]. Our contribution is the quantitative comparison under these explicit controls, rather than a new distance metric or filtering method. The exact nearest-neighbour increment probability mass function (pmf) also shows which morphology information a local distance retains. Our conclusions concern the tested finite windows, not an inferred critical point or a reconstruction of historical implementations.

II. MODEL AND NOISE PROCESSES

A. Parallel-update ballistic deposition

We use a periodic lattice, a flat initial condition and the parallel update

$$h(x,t+1) = \max\{h(x,t)+\eta'_\theta(x,t), h(x-1,t), h(x+1,t)\}, \tag{3}$$

where $\eta'_\theta = \mathbf{1}[\eta_\theta > 0]$ is binary deposition. This model belongs to the family studied with temporally correlated driving [5, 6, 9]. Latent sequences are independent between sites. The spectral inputs are Gaussian; the uniform-driven AR input below is not assumed to be Gaussian.

Equation (3) preserves $|h(x+1,t) - h(x,t)| \le 1$ from a flat state. Each site rises by zero or one per sweep, but can rise because of a higher neighbour even when its own deposition variable is zero. The rule does not reduce to independent deposition (Appendix A). Periodicity gives $\sum_x \Delta h = 0$, so up- and down-step counts are equal in every configuration. With $p_{\text{flat}} = P(\Delta h = 0)$, the increment pmf is

$$P(\Delta h) = \left(\frac{1-p_{\text{flat}}}{2}, p_{\text{flat}}, \frac{1-p_{\text{flat}}}{2}\right). \tag{4}$$

This is a local descriptor, not an assumed order parameter. Different spatial arrangements of the same steps can produce different large-scale interfaces.

B. Shared-innovation AR sum

The streaming construction examined here uses geometrically spaced modes,

$$u_n = aB^{-n}, \quad r_n = e^{-u_n}, \quad (a,B) = (6,2), \tag{5}$$

$$X_n(t) = r_n X_n(t-1) + z(t), \qquad \eta_\theta(t) = \sum_{n=1}^{N} W_n X_n(t), \tag{6}$$

$$W_n^2 = (1-r_n^2)K_\theta u_n^{1-2\theta}, \quad K_\theta = \frac{B^{1/2-\theta} - B^{\theta-1/2}}{\Gamma(2-2\theta)}. \tag{7}$$

Centred innovations $z(t) \sim U[-1/2, 1/2]$ are independent across time and sites but shared by all modes at a given site and time. Modes start at zero, with $N = \max[16, \lceil \log_B(aT) \rceil + 4]$. A common weight rescaling changes raw variance but not the thresholded sequence. At exactly $\theta = 0$ the simulator uses a separate i.i.d. binary reference. The unbranched AR formula remains correlated there.

We call this the *shared AR* variant. Its weights are related to the fast fractional Gaussian noise family [10], but a family name or a symbol $z_t(u_n)$ does not establish innovation covariance across modes. We do not assume every cited simulation uses these streams, initialisation or white-noise shortcut.

Writing $Q_{nm} = \langle z_n(t) z_m(t) \rangle$, the exact stationary finite-mode covariance is

$$C_N(\tau) = \sum_{n,m} W_n W_m r_n^\tau \frac{Q_{nm}}{1 - r_n r_m}, \quad \tau \ge 0. \tag{8}$$

Shared innovations have $Q_{nm} = 1/12$ for all pairs; independent innovations have $Q_{nm} = \delta_{nm}/12$, reducing Eq. (8) to a diagonal sum. These define different processes. Both are compared analytically, while the production AR campaign uses the shared case only. Appendix B addresses cold starts and the interpretation of a power-law approximation.

C. Gaussian spectral filters

For each run and site we filter a length-T Gaussian white sequence in Fourier space, then threshold it. The power-law input uses

$$S_{\text{power}}(f) = |f|^{-2\theta}, \qquad f \ne 0. \tag{9}$$

The second input uses the fGn spectral shape,

$$S_{\text{fGn}}(f) \propto (1 - \cos 2\pi f) \sum_{k=-\infty}^{\infty} |2\pi(f+k)|^{-(2H+1)}, \qquad H = \theta + \tfrac{1}{2}. \tag{10}$$

The alias sum is evaluated for $k = -40, \ldots, 40$. Both spectra are sampled on a finite-record Fourier grid and have DC removed. They are finite-period approximations with a particular low-frequency cutoff, not exact infinite-fGn samples. At $\theta = 0$ the power-law spectrum is flat on nonzero modes, but DC removal still gives a finite-record anticorrelation. The truncated fGn spectrum adds a small approximation. We use *fGn-spectrum filter* with these limitations throughout.

The assembled field has shape (run, t, x): one filtered sequence occupies the complete time column of each site. A memory reshape does not exchange the physical site and time axes. New spectral archives carry an explicit corrected-layout version. Earlier unversioned spectral archives are excluded from all results below.

III. VALIDATION AND STATISTICAL PROTOCOL

A. Validation of the production field

For a centred jointly Gaussian pair with normalised covariance ρ, the centred sign field $s = 2\eta'_\theta - 1$ satisfies

$$C_s(\tau) = \langle s(t+\tau)s(t) \rangle = \frac{2}{\pi} \arcsin \rho(\tau). \tag{11}$$

The finite-period target ρ is computed by inverse Fourier transforming the actual sampled spectrum, including DC removal and alias truncation. It differs from the infinite-fGn target. Equation (11) is not imposed as an identity on the uniform-driven AR input.

The production interface is tested at $\theta = 0.25, 0.40$, $L = 32, 64$ and $T = 65536$, with eight independent runs per filter and parameter pair. Temporal products are circular to match the finite-period target; spatial products test independence at nonzero lags. Errors are standard errors across the eight run estimates. Predeclared numerical tolerances are five run standard errors plus 0.003 per reported correlation. These are engineering checks, not simultaneous confidence statements. Regression tests also cover chunk boundaries, repeated seeds and archive-version preservation.

For a negative control, the same correctly generated field is transposed to (run, x, t) and reshaped back to (run, T, L) without the proper axis mapping. This reproduces a layout fault identified during the implementation audit. It retains filtered sequences but assigns correlations to the wrong coordinates. No production result below uses this control.

B. Simulation grid and late-time width

Every input uses 64 independent runs at each combination of

$$L = 32, 64, 128, 256, 512, \tag{12}$$

$$\theta = 0, 0.10, 1/6, 0.20, 0.25, 0.30, 0.40. \tag{13}$$

There are 105 cells and 6720 runs. The AR duration is $T = \text{round}(15L^{3/2})$; spectral durations round this upwards to a 5-smooth FFT length. At $L = 32$ these are 2715 and 2880 sweeps, respectively; at $L = 512$ they are 173779 and 174960. Thus run counts, sizes and nominal duration factors are matched, with exact durations retained in each archive. Distinct runs and sites have independent streams. The two spectral filters share Gaussian base streams within a parameter cell, so their agreement is not treated as statistically independent replication.

Each run stores 32 geometric-time profiles and a regularly sampled width series $W_r^2(t) = L^{-1}\sum_x [h_r(x,t) - \bar{h}_r(t)]^2$. We average squared widths within each run over the last 20% of stored width samples and then over runs,

$$\overline{W}(L;\theta,T) = \left[\frac{1}{R}\sum_{r=1}^{R}\frac{1}{|\mathcal{T}|}\sum_{t\in\mathcal{T}} W_r^2(t)\right]^{1/2}. \tag{14}$$

The width-sampling stride is $\max(1, \lfloor T/20000 \rfloor)$ sweeps; geometric sampling applies only to profile snapshots. If n_W widths are stored, the terminal mean uses $m = \max(1, \lfloor 0.2 n_W \rfloor)$ samples, and the preceding mean uses the previous m samples. Thus these windows approximate the final and preceding fifths of elapsed time, up to stride and integer rounding. For example, the fGn-spectrum cell at $L = 256$ uses sweeps 49155–61440 for its terminal window, with a three-sweep stride. The five-size slope of $\ln \overline{W}$ against $\ln L$ is called α_{fit}. The adjacent-size slope is

$$\alpha_{\text{eff}}(L, 2L) = \frac{\ln[\overline{W}(2L)/\overline{W}(L)]}{\ln 2}. \tag{15}$$

Neither is assumed asymptotic. We compare the final 20% time window with the preceding 20% as a finite-time diagnostic rather than declaring saturation from duration alone. Widths, flat-pair fractions, window ratios and size slopes use bootstrap standard deviations from 500 resamples of independent runs. Whole run records remain together; different sizes are resampled independently. All adjacent-size estimates use the same resulting bootstrap draws. Five-point fit residuals are not substituted for run-to-run uncertainty.

C. Local distances and sampling uncertainty

We compare exact increment pmfs without rescaling their support or introducing histogram bins. The Hellinger distance is [11, 12]

$$H(P,Q) = \left[1 - \sum_j \sqrt{P_j Q_j}\right]^{1/2}. \tag{16}$$

For Eq. (4),

$$H^2 = 1 - \sqrt{p_\theta p_0} - \sqrt{(1-p_\theta)(1-p_0)}. \tag{17}$$

It contains only the flat-pair fraction and does not measure spatial arrangements or establish a universality class. For $p_\theta = p_0 + \delta p$ with $0 < p_0 < 1$, expansion gives

$$H^2 = \frac{(\delta p)^2}{8p_0(1-p_0)} + O((\delta p)^3). \tag{18}$$

This relation explains the local probability sensitivity of the distance; the distance is not an independent spatial morphology observable.

At each size and for each input, the zero-parameter ensemble is the reference. Equal halves of independent runs give the signal comparison; two disjoint reference halves give its null. The excess is

$$\Delta H = \langle H(P_{\theta,\text{half}}, P_{0,a})\rangle - \langle H(P_{0,a}, P_{0,b})\rangle, \tag{19}$$

averaged over 120 partitions. An outer bootstrap of 200 independent-run resamples repeats this complete estimator, preserving the shared reference within each resample. Its standard deviation supplies the errors. The standard deviation across conditional null partitions is a separate quantity. We define $\Delta H(0, L) = 0$ as the reference point

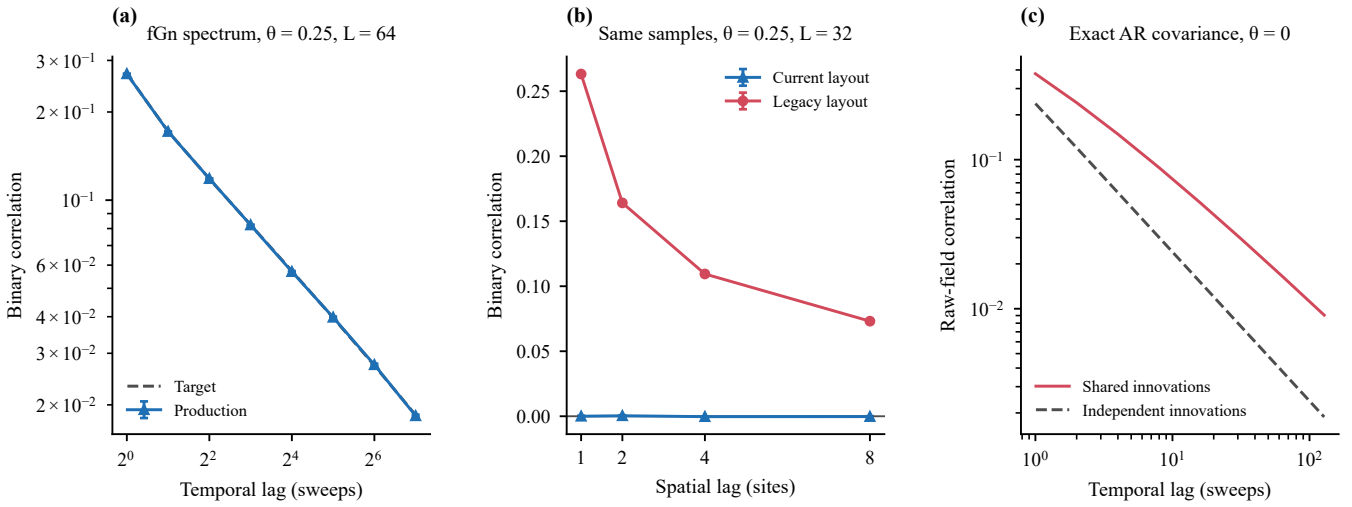


FIG. 1. Input validation. (a) Production binary temporal correlations for the fGn-spectrum filter at $\theta = 0.25$, $L = 64$, $T = 65536$ against its finite-period target. Points are eight-run means with one standard error. (b) Spatial correlations of the same samples at $L = 32$ under correct assignment and the axis-permuted negative control. This is the present implementation's fault control, not an attribution to historical codes. (c) Exact stationary normalised raw covariances of shared- and independently driven AR sums at $\theta = 0$, $N = 48$. No Gaussian binary conversion is applied; these are not the simulator's i.i.d. zero-parameter branch.

instead of treating a self-comparison as an independent signal experiment.

We do not infer a physical onset from the first significance crossing. A shrinking null floor can shift a crossing even when local probabilities are size independent. Conversely, local probabilities can remain fixed while spatial arrangements reorganise. Appendix C supplies a direct counterexample. No physical conclusion here relies on a novelty detector or nominal distance z scores.

IV. RESULTS

A. Spatial and temporal input checks

All eight selected spectral validation cells satisfy the stated checks at the reported lags. At $\theta = 0.25$, $L = 32$, the fGn-spectrum field has binary lag-one temporal correlation 0.2704 against finite-period target 0.2702, with spatial lag-one correlation consistent with zero. The same samples under axis permutation instead give temporal correlation 0.0400 and spatial correlation 0.2632. Figure 1(a,b) illustrates why testing the assembled field is necessary even after validating the one-dimensional filter.

Innovation covariance also matters. At nominal $\theta = 0$, the stationary $N = 48$ AR sums have $C(1)/C(0) = 0.377315$ with shared innovations and 0.237958 with independent innovations. For the shared case, $\tau C(\tau)/C(0)$ rises from 1.04687 at $\tau = 64$ to 1.72174 at 4096; a constant-amplitude $1/\tau$ tail does not describe that window. These raw-field observations do not assign an RG phase to the binary growth model.

TABLE I. Five-size late-time width slopes. Every cell has 64 independent runs; errors are one run-bootstrap standard deviation from 500 resamples. Shared AR includes its i.i.d. shortcut at $\theta = 0$.

θ	Shared AR	Power law	fGn spectrum
0.00	0.503 ± 0.005	0.506 ± 0.005	0.506 ± 0.005
0.10	0.606 ± 0.006	0.542 ± 0.007	0.545 ± 0.007
1/6	0.641 ± 0.008	0.584 ± 0.006	0.588 ± 0.006
0.20	0.657 ± 0.008	0.618 ± 0.007	0.621 ± 0.007
0.25	0.717 ± 0.008	0.682 ± 0.007	0.678 ± 0.007
0.30	0.761 ± 0.008	0.740 ± 0.007	0.745 ± 0.007
0.40	0.845 ± 0.010	0.836 ± 0.009	0.833 ± 0.009

B. Increasing finite-size width slopes

The white-noise references give $\alpha_{\mathrm{fit}} = 0.503 \pm 0.005$, 0.506 ± 0.005 and 0.506 ± 0.005 for shared AR, power law and the fGn-spectrum input, close to 1/2 over this window. The width slopes rise with θ in all three campaigns (Fig. 2, Table I). Spatially independent spectral inputs do not produce an approximately flat 1/2 branch over this finite-size range.

At $\theta = 0.25$, shared AR and the fGn-spectrum filter give 0.717 ± 0.008 and 0.678 ± 0.007; at 0.40 they

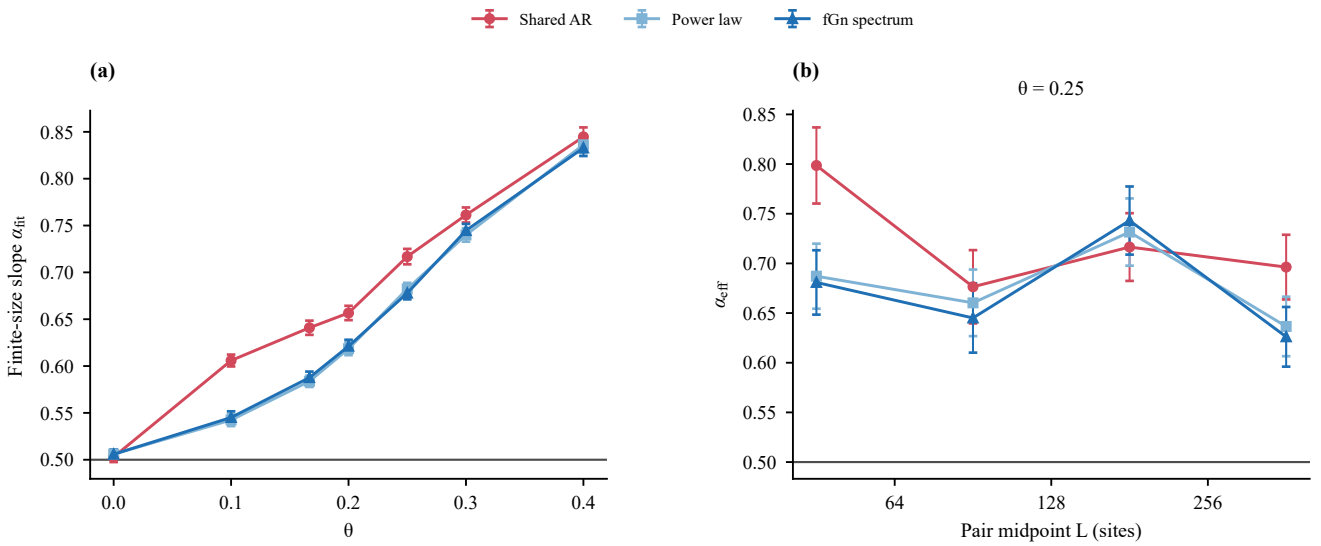


FIG. 2. Regenerated width slopes. (a) $\alpha_{\rm fit}$ with one run-bootstrap standard deviation. The horizontal 1/2 line is a reference, not a fitted phase boundary. (b) Adjacent-size $\alpha_{\rm eff}$ at $\theta = 0.25$, with bootstrap errors, at geometric size-pair midpoints. All four pairs are shown. These finite-window quantities do not measure asymptotic exponents or θ_c.

give 0.845 ± 0.010 and 0.833 ± 0.009. Thus construction changes finite-size amplitudes and slopes, but the large rise with temporal correlation also occurs under both Gaussian inputs. The maximum difference between the two spectral-filter slopes is 0.0047. Their base streams are matched, and their agreement is not independent confirmation.

Adjacent-size estimates fluctuate rather than show a monotone flow. At $\theta = 0.25$ the fGn-spectrum sequence is approximately 0.681, 0.645, 0.743, 0.626, with standard deviations about 0.030–0.035. This does not establish convergence to 1/2 or a long-range fixed point. Across all cells, the final-to-preceding-window RMS width ratio lies between 0.935 and 1.053. This checks late-time stability without proving that every finite-time correction has vanished; duration, cutoff and initialisation remain part of the protocol.

C. Paired filter comparison and duration controls

For the Gaussian filters, corresponding run records are paired by seed at each size and parameter. We resample the 64 pairs jointly within each cell, independently between sizes, and refit the difference $\delta\alpha = \alpha_{\rm fGn} - \alpha_{\rm power}$ in each of 2000 bootstrap replicates. Pointwise 95% percentile intervals contain zero at all seven parameters (Table II). At $\theta = 0.25$, $\delta\alpha = -0.00443$ with interval $[-0.01104, 0.00219]$. Thus the present ensembles do not resolve a difference between these width slopes; this is neither an equivalence test with a prespecified margin nor independent replication of a scaling law.

To separate observation duration from changes in spectral cutoff, we also generate one length-$4T$ fGn-spectrum record per run at each combination of $\theta = 0.10, 0.40$ and $L = 128, 256$. Here T is the original spectral campaign duration for that size. Each of 32 new independent runs starts from a flat interface and uses nested prefixes of the same record to reach T, $2T$ and $4T$. The terminal fifth of regularly sampled squared widths defines each estimate. Width ratios use 2000 paired run-bootstrap resamples. The cutoff remains $1/(4T)$ throughout a cell, so its T prefix is a separate control, not a repeat of the original length-T periodic input. Figure 3 reports the resulting ratios and uncertainties. At $4T$, the four central ratios range from 0.989 to 1.034. For $L = 256$, they are 1.031 ± 0.025 at $\theta = 0.10$ and 1.034 ± 0.033 at 0.40, where errors are one bootstrap standard deviation. All eight pointwise 95% ratio intervals contain unity. No duration-induced width change is resolved at this precision in these controls; this does not demonstrate saturation or exclude smaller drifts. These controls test observation-time sensitivity under a fixed input record; they do not establish

TABLE II. Paired Gaussian-filter slope differences and pointwise 95% run-bootstrap percentile intervals. Sizes and central estimators are the same as in Table I.

θ	$\delta\alpha$	95% interval
0.00	−0.00010	$[-0.00240, 0.00220]$
0.10	0.00266	$[-0.00366, 0.00932]$
1/6	0.00390	$[-0.00289, 0.01081]$
0.20	0.00269	$[-0.00510, 0.01010]$
0.25	−0.00443	$[-0.01104, 0.00219]$
0.30	0.00468	$[-0.00100, 0.01037]$
0.40	−0.00333	$[-0.00869, 0.00184]$

convergence with respect to the spectral cutoff.

D. Local morphology and detection floors

At $L = 256$, the flat-pair fraction decreases as correlations strengthen for every input [Fig. 4(a)]. For the fGn-spectrum filter it changes from 0.4364 ± 0.0044 at $\theta = 0$ to 0.1481 ± 0.0044 at 0.40; shared AR changes from 0.4233 ± 0.0042 to 0.1138 ± 0.0035. There is no opposite-sign response between the correctly assembled inputs on this grid. At $\theta = 0.40$, $L = 256$, the shared-AR and fGn-spectrum late-time widths are 17.97 and 13.80, a ratio of 1.30. Figure 5 compares their final profiles and ensemble spatial spectra. A single size and spectrum are not used to identify a faceting or anomalous-scaling boundary.

The Hellinger excess detects local changes under correlated driving [Fig. 4(b)], but its magnitude and size dependence do not define a unique physical onset. In the fGn-spectrum reference, the conditional null mean changes from 0.0155 at $L = 32$ to 0.0036 at 512, a factor of 4.31, despite matched run counts. Intermediate sizes need not be perfectly monotone. The number of columns and their spatial correlations affect each run's pooled-pmf precision.

A first crossing of a fixed number of null standard deviations mixes local probability changes with detection power. This statement concerns the estimator used here. Without reconstructing another study's observable, samples and decision rule, it does not establish that a reported threshold has the same origin. A detectable local difference likewise does not demonstrate a change of scaling universality.

V. DISCUSSION

The central result is a finite-window comparison in which input validation and statistical comparison are treated separately. Both Gaussian spectral filters yield increasing width slopes with temporal memory, and paired resampling does not resolve their slope differences at the tested parameters. The shared AR variant differs most visibly at weak and intermediate memory. Agreement between the Gaussian filters therefore does not imply insensitivity to every noise construction, while their common trend shows that increasing width slopes are not exclusive to shared AR innovations.

The validation controls identify two distinct requirements. First, correlation measurements must reach the final space–time field passed to the growth rule. An axis-assignment fault can preserve filtered sequences while altering the coordinates in which their correlations act. Second, an AR specification must include the covariance of its innovations across modes. Identical mode weights with shared and independent innovations describe different input processes. Our analytic comparison establishes this distinction; isolating its effect on growth would require a separate matched innovation-control campaign.

The fixed-record controls address observation duration without changing the lowest available input frequency within each cell. Their interpretation is correspondingly restricted to that input record and to the tested sizes and parameters. They do not control the difference between periodic spectral inputs and cold-started AR streams. Residual differences can also involve short-time correlations and non-Gaussian marginals. Matching nominal θ is insufficient to isolate any of these mechanisms.

Local distances answer a different question from spatial scaling. Equation (17) and its expansion show that the increment-pmf Hellinger distance measures changes in one local probability. Its sampling floor affects detection power, but the distance discards the arrangement of steps. The explicit counterexample in Appendix C shows why neither a constant distance nor a detectable departure identifies a scaling universality class. Comparisons with emergence diagnostics [7] must retain the observable and sampling rule of each study.

Microscopic sensitivity under temporal correlations has already been emphasised [8]. The present work adds an explicit field-assembly control, innovation-covariance specification, paired filter inference and fixed-record duration checks in one discrete model. The production grid stops at $L = 512$ and uses durations proportional to $L^{3/2}$; bootstrap errors do not quantify residual cutoff or initialization effects. Establishing asymptotic exponents or a faceting boundary would require larger sizes, independently controlled cutoffs and stationary input comparisons. The different threshold definitions in the literature therefore remain separate physical questions, and no historical threshold is invalidated here.

VI. CONCLUSION

Specifying the complete noise process and observation protocol makes the finite-window comparison interpretable. Correctly assembled Gaussian spectral inputs and the shared AR variant all show increasing width slopes and decreasing flat-pair fractions as temporal correlations strengthen. Paired run-bootstrap inference resolves the uncertainty in the Gaussian-filter comparison, and fixed-record controls test duration sensitivity separately from a changing spectral cutoff. Exact innovation covariances and the local-pmf reduction explain why generator labels and local distance thresholds cannot substitute for these checks. The resulting benchmark supports finite-window comparisons without identifying them with asymptotic exponents or a measured critical temporal parameter.

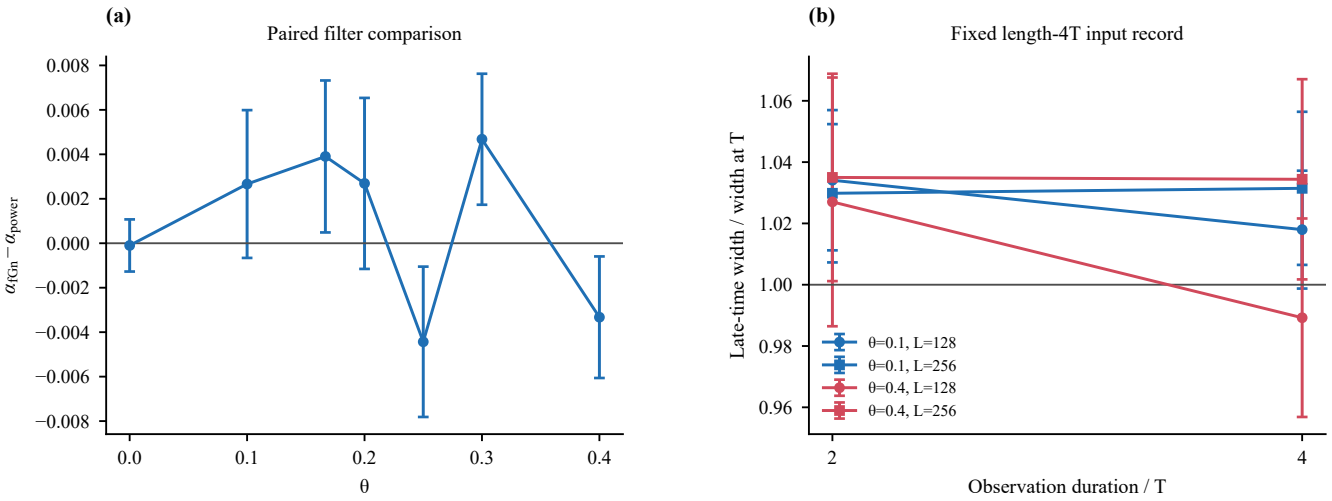


FIG. 3. Additional inference and duration controls. (a) Paired Gaussian-filter slope differences with one bootstrap standard deviation; Table II reports percentile intervals. (b) Late-time RMS width at $2T$ and $4T$, divided by its value at T, using nested prefixes of one fixed length-$4T$ fGn-spectrum record per run. Colours distinguish θ and markers distinguish sizes. Errors are one paired run-bootstrap standard deviation from 32 independent runs per cell.

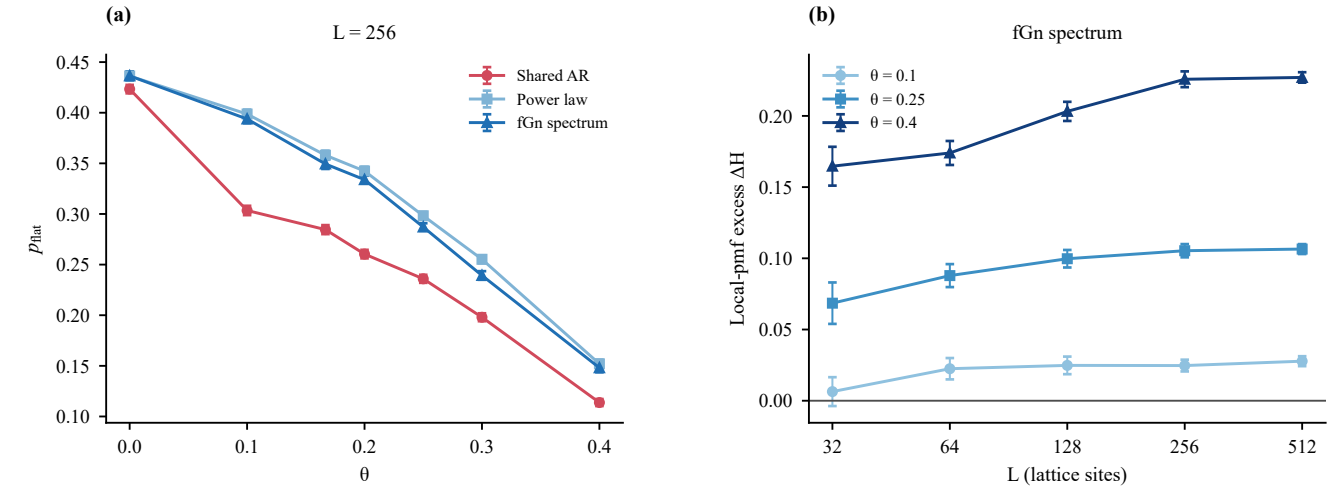


FIG. 4. Local morphology and pmf distances. (a) Flat-pair fraction at $L = 256$ from all 64 final profiles per cell, with run-bootstrap standard deviations. Every input shows a decrease on this grid. (b) Excess Hellinger distance in the fGn-spectrum campaign at three fixed θ values, with outer independent-run bootstrap standard deviations from 200 resamples. Conditional partition scatter is not substituted for these errors. This local statistic is not a necessary or sufficient test for a universality crossover.

DATA AVAILABILITY

The code, production data and additional control results are maintained in `https://github.com/HuangYong2006/bd-correlated-noise-1.0.0-code`. The reproducibility materials contain all 105 production cells, validation outputs, analysis and plotting scripts, seeds, layout identifiers and source hashes. The reproduction guide maps figures and tables to their inputs.

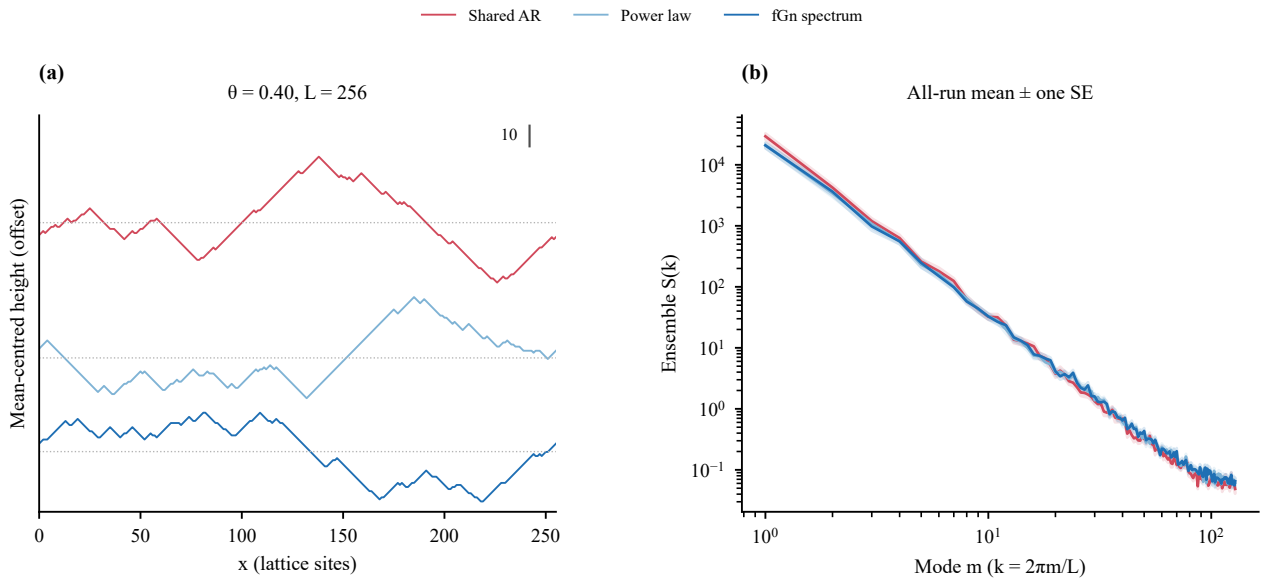


FIG. 5. Final interfaces and spectra at $\theta = 0.40$, $L = 256$. (a) One illustrative profile per input, selected by final width nearest the 64-run median. Profiles are mean-centred and vertically translated without amplitude rescaling. Dotted lines show baselines; the bar denotes ten height units. (b) Mean $S(k) = L^{-1}|\hat{h}(k)|^2$ over all 64 final profiles, with one run standard error shaded. Every positive Fourier mode is shown without smoothing or an artificial floor. No asymptotic spectral exponent is fitted from this panel.

Appendix A: Single-step constraint

If all neighbour differences have magnitude at most one, Eq. (3) gives $h(x,t) \le h(x,t+1) \le h(x,t)+1$. The maximum at the next site includes $h(x,t)$, so $h(x+1,t+1) \ge h(x,t)$ and $h(x,t+1) - h(x+1,t+1) \le 1$. Exchanging sites gives the reverse bound. Induction from the flat state proves the constraint. Periodic summation forces equal up- and down-step counts. No height-inversion symmetry of the maximum update is needed.

Appendix B: Covariance and limiting behaviour

Stationarity gives $\Sigma_{nm} = r_n r_m \Sigma_{nm} + Q_{nm}$, hence $\Sigma_{nm} = Q_{nm}/(1 - r_n r_m)$ and Eq. (8). With zero initial modes and t completed innovations, the covariance instead is

$$C_N(t,\tau) = \sum_{n,m} W_n W_m r_n^{\tau} Q_{nm} \frac{1-(r_n r_m)^t}{1 - r_n r_m}. \tag{B1}$$

A finite cold-start record is not silently identified with a stationary one. The stationary shared-innovation spectrum is

$$S_N^{\mathrm{shared}}(\omega) = \sigma_z^2 \left| \sum_n \frac{W_n}{1 - r_n e^{-i\omega}} \right|^2, \tag{B2}$$

whereas independent innovations give a sum of squared transfer functions. Cross terms cannot be discarded in the shared case.

The impulse response $g_j = \sum_n W_n r_n^j$ has continuum-grid approximation $g_j \propto j^{\theta-1}$ for intermediate $u_N j \ll 1 \ll u_1 j$. At $\theta = 0$, a $1/j$ kernel combined with

$$\sum_{j=1}^{\infty} \frac{1}{j(j+\tau)} = \frac{H_\tau}{\tau}, \qquad H_\tau \simeq \ln \tau + \gamma, \tag{B3}$$

permits a logarithmic correction. This mechanism argument is not an exact finite-N pure-power asymptote: every finite-mode covariance ultimately decays exponentially. Convergence at fixed lags does not exchange infinite-mode and infinite-lag limits.

Appendix C: Information missing from a local pmf

On an even periodic lattice, consider equal counts of $+1$ and -1 increments without zeros. Alternating signs and a block of $L/2$ positive followed by $L/2$ negative increments have identical nearest-neighbour pmfs. Yet the alternating interface has bounded width and the block arrangement has width proportional to L. These are information-theoretic examples, not assertions about stationary states of Eq. (3). They show why the single-step constraint does not make a local pmf sufficient for large-scale roughness. Constant ΔH cannot rule out a crossover, and increasing ΔH cannot alone establish one.

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